

3.4 Optional application 2 – statistics

3.4.1 SA: Discrete random variables (DRVs) and expectation

	Content
SA1	Understand DRVs with distributions given in the form of a table or function.
	Content
SA2	Evaluate probabilities for a DRV.
	Content
SA3	Evaluate measures of average and spread for a DRV to include mean, variance, standard deviation, mode and median.
	Content
SA4	Understand expectation and know the formulae: $E(X) = \sum x_i p_i$; $E(X^2) = \sum x_i^2 p_i$; $\text{Var}(X) = E(X^2) - (E(X))^2$
	Content
SA5	Understand expectation of linear functions of DRVs and know the formulae: $E(aX + b) = aE(X) + b$ and $\text{Var}(aX + b) = a^2 \text{Var}(X)$ Know the formula $E(g(X)) = \sum g(x_i) p_i$ Find the mean, variance and standard deviation for functions of a DRV such as $E(5X^3)$, $E(18X^{-3})$, $\text{Var}(6X^{-1})$

	Content
SA7	Proof of mean and variance of discrete uniform distribution.